

Anisur Rahman

Sr. Software Engineer at AppsCode

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TECHNICAL SKILLS

Languages: Go (Golang), C++ , Bash

Platforms: Linux, Docker, Kubernetes, Openshift, Cloud-Native, AWS, GCP

Technologies: REST, gRPC, Protocol Buffers, Makefile, Helm, CI/CD, GitOps, FluxCD, Prometheus, Grafana

Storage systems: LVM, ZFS, K8s-CSI, Restic, Wal-G, Velero, Rook-Ceph, PostgreSQL, MySQL

Core Strengths: Microservice, Distributed Systems, Raft, Concurrency, Parallel Processing, Software Architecture

WORK EXPERIENCE

Senior Software Engineer

AppsCode Ltd, Dhaka, Bangladesh

January 2023 - Present

- **Golang:** Write high-performance Go code for distributed, cloud-native applications.
- **DBaaS:** Developed k8s-native managed database and disaster recovery platforms (KubeDB, KubeStash) for production clusters across global enterprises.
- **Parallel Task Scheduling System:** Built a distributed scheduler for parallel DB backup and restore, delivering 30% faster performance.
- **Multi-Cloud Gateway:** Integrated NooBaa Multi-Cloud Gateway into KubeStash, enabling cross-cloud S3 bucket replication to resolve storage failover.
- **CSI driver:** Extended the Topo-LVM CSI driver to support snapshot chunking and offloading to public object storage for hybrid and on-prem setups.
- **DB Migrator:** Architected and developed a database migration framework for *RDS* ⇌ DBaaS in k8s (PG and MySQL), optimized with parallel processing and snapshot dumping, delivering ultra-fast migrations.
- **SRE:** Ensured reliability of production Kubernetes clusters on bare metal using distributed storage (Rook-Ceph), resolved critical production issues for Orange, Nokia, Huma, and Emerson, ensuring 99.9% uptime.
- **Presentation:** Authored webinars, documentation, and blogs about DBaaS management in k8s. [1], [2], [3], [4]

PROJECTS

KubeStash 🌐 🌐

Go • Kubernetes • Helm • Kubebuilder • Cloud • Concurrency • Restic • Prometheus • Grafana

A complete disaster recovery solution for Databases, Workload applications and Cluster for Kubernetes environment.

- Led development and ownership for 2.5 years from system design to release.
- Added features (retry, validation, and failure recovery) to make it a leading k8s disaster recovery solution.
- Added bucket as backup storage for all major clouds with cloud identity integration, e.g., IRSA, Pod identity.
- Architected sharded operators and controllers for large-scale, parallel backups across approx 1000+ DBaaS.
- Engineered high-performance backup & restore plugins (concurrent, snapshot, PITR) for 10+ databases.
- Wrote Prometheus exporters and designed a Grafana monitoring dashboard.
- Authored automated E2E and chaos tests to validate reliability under 1000 database workloads.

KubeDB WAL-G 🌐 🌐

Go • gRPC • Databases • Distributed System • Chunking

Wal Archival and Restoration for Databases in the Cloud

- Built a WAL archival and restoration framework supporting Point-in-Time Recovery (PITR) for KubeDB databases.
- Implemented continuous WAL shipping to any object storage.
- Architected a gRPC proxy server for WAL streaming in Kubernetes distributed environments.
- Automated DBaaS failover detection and restore from WAL backup.

PROJECTS

TopoLvm CSI Driver

Go • gRPC • Protocol Buffers • LVM • Chunking • Restic • Prometheus • Grafana

Restic-based Volumes snapshot feature for kubernetes PVC

- Architected a gRPC server to take volume snapshot by using Kubernetes CSI workflows.
- Efficient backup and restore operations for persistent volumes for hybrid and on-prem setups.
- Wrote Prometheus exporters for monitoring volume snapshot operations.

Migrator

Go • gRPC • Protocol Buffers • Prometheus

High-performance data migration engine for PostgreSQL & MySQL

- Architect and Developed parallel/concurrent pipelines for high-volume database data transfer.
- Designed a gRPC-based control plane for managing complex migration workflows.
- Achieved approx +-10 hours to migrate 2 TB of MySQL data from RDS to kubedb managed DBaaS.

TaskQueue

Go • Concurrency • Parallel Processing

A Kubernetes controller that manages a queue of distributed task in kubernetes environment

ACCOMPLISHMENTS

Opensource Projects

Contributor to WAL-G, Stash, Go-Shell, and IBM MultiCloud Gateway.

- Multi-writer support for the go-shell. [PR] — External DB secret reconciliation for IBM MCG. [PR]
- HookExecutionPolicy. [PR] — Charset converter to non-UTF-8. [PR] — TLS Verification support. [PR]

Competitive Programming

2000+ problems solved across Codeforces, Atcoder, UVA, Spoj, etc.

- CodeChef Master (Max Rating: 2195), Codeforces Expert (1639)
- Rank 3.1%: ICPC, Dhaka Regional 2022. [DIU-Sedulous], Rank 3.4%: BUET IUPC Contest 2022. [DIU-Sedulous]

Project 4 title, Sed sagittis consectetur eros, eu auctor felis lobortis vel. Sed eu risus at nunc viverra finibus. Curabitur eu quam libero. Sed euismod ante in tellus efficitur, sit amet accumsan mi sodales. [Try it!](#)

EDUCATION

Daffodil International University

BSc in Computer Science and Engineering

CGPA: 3.87 Out of 4.00

Dhaka, Bangladesh

Jan 2019 - Dec 2022